COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure & Algorithms	**SECTION:**	CPE21S4
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NAME:	Braganza, Ralph Bal, Kevin Ranz J. Llaneta, Jhunel Uchi, Mao Sophie Verano, James Daniel	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

Seatwork 8.1 Using Sorting Algorithms 2

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using sorting algorithms.

1. **To be familiarized with the sorting algorithms.**
2. **To be able to different the types of sorting algorithms.**
3. **To be able to create a program with sorting algorithms.**

Answer the following questions:

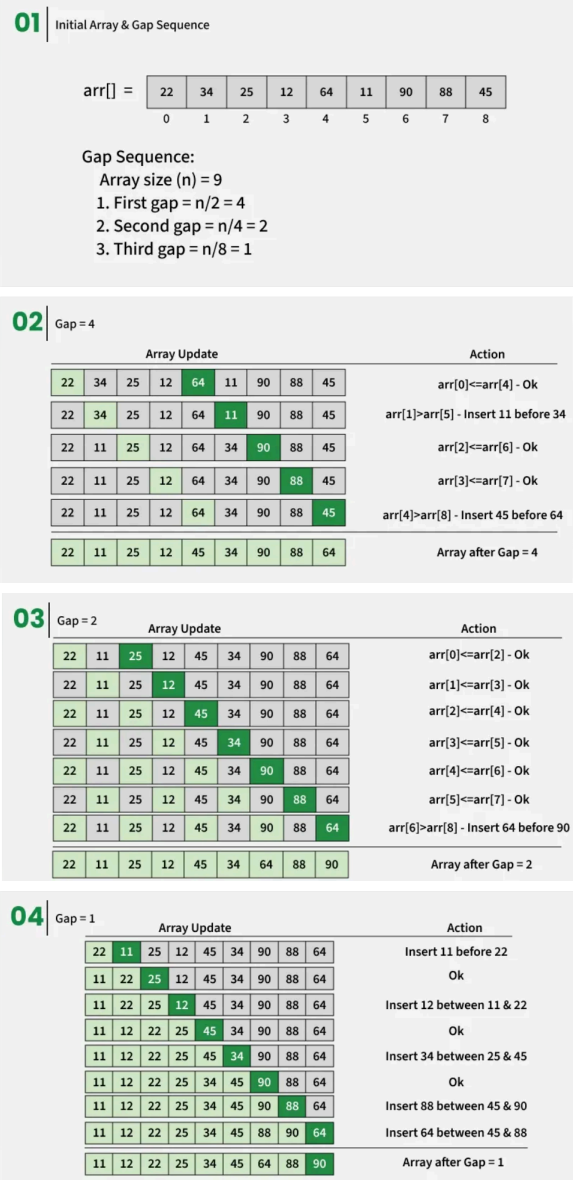
1. Explain the quick sort, shell sort and merge sort types of sorting algorithms.

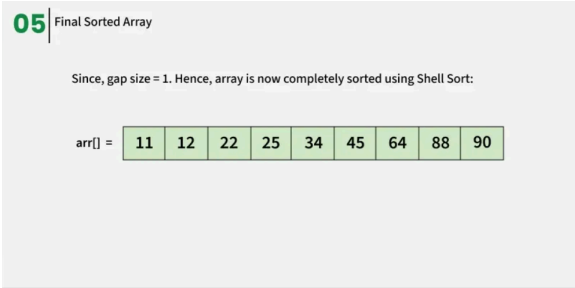
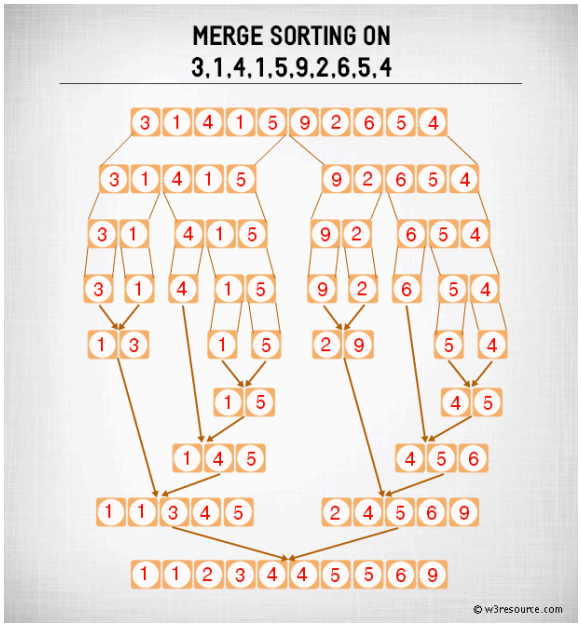
Shell Sort:
This divides the total size of the array by a specific number to get a starting value. If this value has a decimal, the decimal is ignored, and the remaining whole number is called the "gap." This gap determines how many indexes the algorithm will jump over to compare, swap, and sort values. This process repeats, shrinking the gap each time. Once the gap value becomes 1, it applies a standard insertion sort to compare, swap, and fully sort the remaining values.

Merge Sort:
This sort splits the array down the middle into a "left" array and a "right" array. It keeps splitting these arrays down until they only have two values, and then splits them again into individual, single values. Next, it compares these separate values, putting the lowest value in the front and the highest value in the back. It does this for both the left and right sides. After finishing both sides, it combines them back together. Since they are still not fully sorted, it does one final process which is comparing, sorting, and swapping until the entire array is sorted.

Quick Sort: This sort divides the data around a single element called the “pivot”, which is placed on the far right at the last index. The algorithm checks each value in the array. If the value is greater than the pivot **(if(j > pivot))**, it just moves to the next index. If the pivot is greater than the index value **(if(j < pivot))**, it moves the temporary index to the right and swaps the values. After that, it moves the main index to the right.The last condition is, if the value is equal to the pivot **(if(j == pivot))**, the temporary index value is swapped with the pivot value. This loop continues until the sorting is complete.

2. Give simple sample programs in C++ that uses the above sorting algorithms. Use user input of 10 integer values as the array elements to be sorted. Explain how the programs work.

Sorting Algorithms	VISUAL REPRESENTATION	Explanation
[SHELL SORT]		Shell sort algorithm is known as an in-place comparison and an optimization of **insertion sort**. **(geeksforgeeks, 2025)** It uses a gap sequence to compare two numbers which allows elements to move over large distances which reduces the number of swaps required as numbers are being compared between gaps. After comparing elements that are far apart together, the gap gradually reduces as each comparison and iteration sorts across the element array which increases the efficiency. In the very end where the gap has been reduced to 1, it is just using the insertion sort to fix the remaining unsorted values.

		
[MERGE SORT]		Merge Sort is a sorting algorithm that works by dividing an array into smaller parts, sorting those parts, and then combining them back together in the correct order. This algorithm uses divide and conquer to be able to sort the given values **(GeeksforGeeks, 2026)**. It divides by splitting the array into two smaller parts repeatedly until each part has only one element in the array. Conquer treats these elements as sorted and compares them with other elements as the smaller groups are formed. Finally, we have merge, where it combines the sorted groups together while arranging the values from smallest to largest until the entire array is sorted. In short, it works by splitting, sorting and combining the values of the array to have a sorted array of values at the end.
[QUICK SORT]		Quick Sort algorithm is a based division sorting algorithm that picks an element as a pivot and partitions the given array around the picked pivot. **(geeksforgeeks, 2014)**, The algorithm chooses pivot, the pivot is served as the element to be compared to throughout the array. The quick sort is partitioned into smaller parts. In partitioning an array, the elements smaller than the pivot will be on its left, and all greater than will be on its right. This sorting algorithm has a best-time

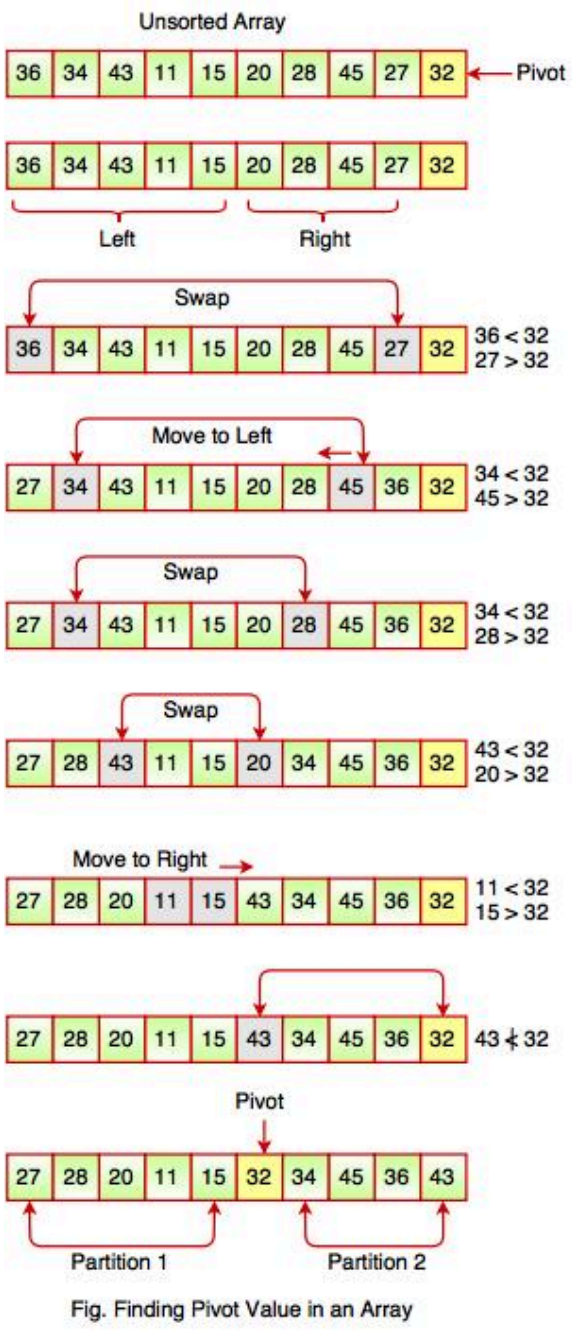


Fig. Finding Pivot Value in an Array

complexity case of O(n log n) and a worst case of O(n2). In choosing the pivot, the representation shows that in the partitioned array, the pivot is always the last element. In sorting the algorithm, the pivot and the j are compared as the array iterates. In the process of comparison, there are 3 conditions that are based on the process of sorting.

***C1**: if j > pivot, increment j*

***C2**: if the j < pivot, increment i and swap i and j. Increment j*

***C3**: if j == pivot increment i then swap to pivot*

And the sorting keeps comparing and processing conditions until the element array is sorted.

REFERENCES:

GeeksforGeeks. (2025, December 20). *Shell Sort*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/shell-sort/

GeeksforGeeks. (2026, August 6). *Merge sort*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/merge-sort/

GeeksforGeeks. (2014, January 7). *Quick Sort*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/quick-sort-algorithm/

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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